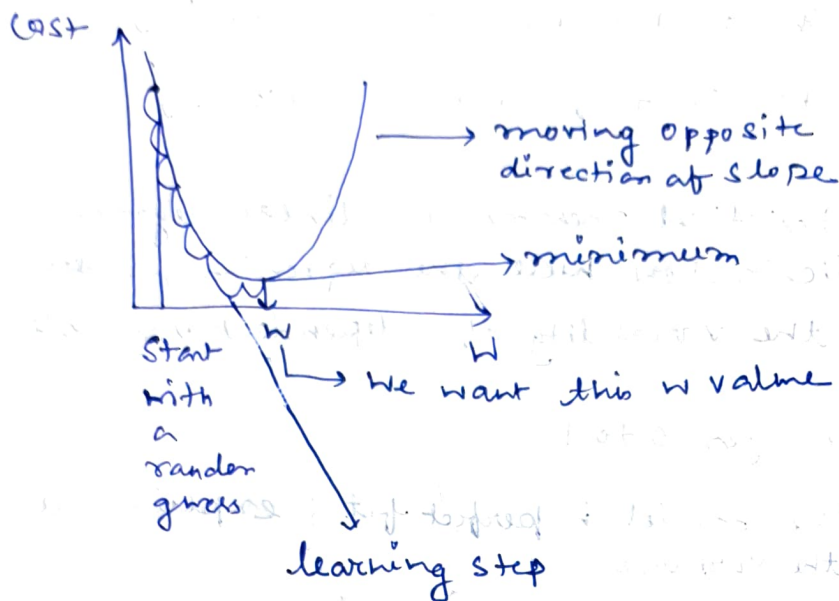


## Iterative optimization

- Start with a random guess for  $w$ .
- iteratively update it to minimize cost function.



## Gradient descent approach for iterative optimization

GD is an optimization algorithm used to minimize a function by iteratively moving towards the direction of steepest descent (the negative gradient).

Here, we use for minimizing  $J(w)$

- Imagine you are standing on a mountain and want to reach bottom.
- At each step, look the slope (gradient) and move downhill.
- If step is too big, might overshoot.
- If step is too small, take too long.

Partial derivatives give us the slope (i.e. direction to move) with respect to a parameter.

## Multiple linear regression

$$y = XW + \epsilon$$

$m$  = no. of training samples

$n$  = no. of features.

$$\text{Model: } \hat{y} = XW$$

$$X_i = [x_{i,0} \quad x_{i,1} \quad \dots \quad x_{i,n}]$$

$$X = \begin{bmatrix} 1 & x_{1,1} & x_{1,2} & \dots & x_{1,n} \\ 1 & x_{2,1} & x_{2,2} & \dots & x_{2,n} \\ \vdots & \vdots & \vdots & \ddots & \vdots \\ 1 & x_{m,1} & x_{m,2} & \dots & x_{m,n} \end{bmatrix}$$

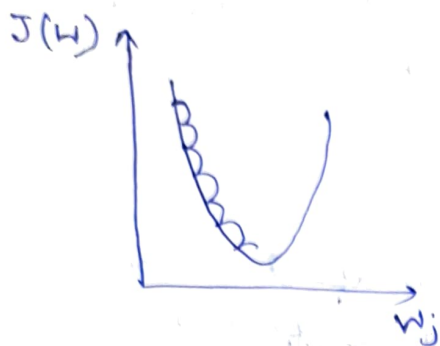
$$\hat{y}_i = \sum_{j=0}^n w_j x_{i,j}$$

$$x_{i,0} = 1$$

$$J(W) = \frac{1}{2m} \sum_{i=1}^m (y_i - \hat{y}_i)^2 \quad \text{-- MSE / RSS}$$

Try to estimate  $\hat{W}$  by minimizing  $J(W)$

- It is a convex error function
- Geometrically, error surface is bowl shaped.



Simultaneously  
update

$j = 1, 2, 3, \dots, n$

$$w_j^{itr+1} = w_j^{itr} - \alpha \frac{\partial}{\partial w_j} J(W)$$

learning rate 0 to 1

$$\begin{aligned} \frac{\partial}{\partial w_j} J(W) &= \frac{\partial}{\partial w_j} \left[ \frac{1}{2m} \sum_{i=1}^m (y_i - \hat{y}_i)^2 \right] \\ &= \frac{1}{m} \sum_{i=1}^m (y_i - \hat{y}_i) (-x_{i,j}) \end{aligned}$$

$$w_j^{itr+1} = w_j^{itr} - \alpha \frac{1}{m} \sum_{i=1}^m (\hat{y}_i - y_i) x_{i,j}$$

epoch: training with all samples.

### Algorithmic representation of GD

Step 1: Randomly initialize the parameters  $w_0, w_1, \dots, w_n$  and learning rate.

Step 2: Calculate  $\hat{y}_i$  for all training samples

Step 3: Calculate  $J^{itr}(W)$  for  $itr^{th}$  epoch

Step 4: Update all parameter  $w_0, w_1, \dots, w_n$

Step 5: Repeat step 2-4 until  
 $|J^{itr}(W) - J^{itr-1}(W)| > \rho$

$\rho$ : Small value.

## Variant of GD

### Batch gradient descent

- All training samples are taken into consideration for single updating of all the parameters.
- problem with huge training sample.

### Stochastic gradient descent

- Update parameter by considering just one training sample at a time (repeat epoch by epoch till convergence).

$$w_j^{itr+1} = w_j^{itr} + \alpha (y_i - \hat{y}_i) x_{ij}$$

### Mini-batch GD

- Use small batch of training samples for updation.
- Difference between all training variant of GD. (Self Study)

#### Parameter

Model parameter that can be estimated by fitting the given data

$w_0, w_1, \dots, w_n$

#### hyperparameter

→ Values set before the model start training

→ Can not be learned by fitting the model.

$\alpha, \rho$ , number of epochs.